



Core Concepts

- **Image:** Read-only template (OS + app + deps)
- **Container:** Running instance of an image
- **Docker Daemon:** Background service
Handles image pulls, container lifecycle, scaling, networking
- **Docker Desktop:** GUI + VM managing Docker daemon & machine state

Image → Container Relationship

- **1 image → many containers**
- Containers are isolated but share the same image layers

Basic Commands

Run Container

```
docker run -it ubuntu
```

- **i** interactive
- **t** terminal

```
docker run calcom
```

List Containers

```
docker container ls# running  
docker container ls -a# all
```

Stop / Start

```
docker stop <container_name>  
docker start <container_name>
```

Execute Command in Running Container

```
dockerexec <container_name> ls  
dockerexec -it <container_name> bash
```

Images

List Images

```
docker image ls
```

Pull Image

```
docker pull ubuntu
```

Port Mapping

```
docker run -it -p 1025:1025 <image_name>
```

- `host_port : container_port`

Environment Variables

```
docker run -e ENV=prod -e DB_HOST=localhost ubuntu
```

or

```
docker run --env-file .env ubuntu
```

Dockerfile (Dockerizing an App)

Basic Structure

```
FROM ubuntu

RUN apt update && apt install -y curl

COPY . /app
WORKDIR /app

RUN make build

ENTRYPOINT ["/app"]
```

Key Instructions

- **FROM** → base image
- **RUN** → execute during build
- **COPY** → copy files
- **WORKDIR** → set working dir
- **ENTRYPOINT** → fixed startup command
- **CMD** → default args (overridable)

Build Image

```
docker build -t imagename .
```

Image Caching

- Docker caches **layers**
 - Change in one step invalidates only layers **after it**
 - Put stable steps (apt install) **before** app code copy
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Publishing Images (Docker Hub)

Steps

```
docker login
docker tag imagename username/imagename
docker push username/imagename
```

- Create Docker Hub account first
 - Public images = open source by default
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Docker Compose (Multi-Container Setup)

docker-compose.yml

```
version:"3.8"

services:
  postgres:
    image:postgres:15
    ports:
      -"5432:5432"
    environment:
      POSTGRES_USER:admin
      POSTGRES_PASSWORD:secret
```

Commands

```
docker compose up -d  
docker compose down
```

- `d` → detached mode
 - Automatically creates network
 - One command → entire stack
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Mental Model

- Docker ≠ VM
- Containers **share kernel**
- Lightweight, fast startup
- Perfect for **replication, CI/CD, microservices**